

Analysis on Credit Card Fraud Identification Techniques based on KNN and Outlier Detection

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Abstract—Popular payment mode accepted both offline and online is credit card that provides cashless transaction. It is easy, convenient and trendy to make payments and other transactions. Credit card fraud is also growing along with the development in technology. It can also be said that economic fraud is drastically increasing in the global communication improvement. It is being recorded every year that the loss due to these fraudulent acts is billions of dollars. These activities are carried out so elegantly so it is similar to genuine transactions. Hence simple pattern related techniques and other less complex methods are really not going to work. Having an efficient method of fraud detection has become a need for all banks in order to minimize chaos and bring order in place. There are several techniques like Machine learning, Genetic Programming, fuzzy logic, sequence alignment, etc are used for detecting credit card fraudulent transactions. Along with these techniques, KNN algorithm and outlier detection methods are implemented to optimize the best solution for the fraud detection problem. These approaches are proved to minimize the false alarm rates and increase the fraud detection rate. Any of these methods can be implemented on bank credit card fraud detection system, to detect and prevent the fraudulent transaction.

Keywords: Classification, Fraud Detection, K-Nearest Neighbor Algorithm, Outlier Detection.

I. INTRODUCTION

In day-to-day usage of credit card transactions the procurement of products and services assists online transactions or card swiping procurements. This leads to increase in online transactions using credit and debit cards evolving to a world of effortless expenditure. Frauds involved in the credit card section have caused severe damage to the users and the service provider and is said to be even worse in coming days. Fraudsters observe and adapt to the quick changes in the technology and find clever ways to involve in illegal activities [6]. Frauds caused due to these smart hackers are hazardous and dangerous. A well-educated fraudster can create several identities and conduct credit card transactions without being caught.

Talking in terms of e-commerce transactions the major problem faced due to these fraudulent activities is so similar to legal ones. Hence having an efficient and complex fraud detection system is a must to prevent these fraudulent activities. The challenging section of this problem is to detect frauds in a huge dataset where the legal transactions are more and the fraudulent transactions are bare minimum or close to negligible. There are very few papers on credit card fraud detection

methods due to the fact that these methods cannot be tested without a dataset. Hence it's difficult to prove the robustness or even the probability of success ratio of the methods. As we know that the credit card information is confidential, the bank owners and service providers do not encourage in sharing these data for experiments as well. Hence KNN is used widely for finding the final results obtained in ranking the research methodology. In this paper we investigate credit card fraud using KNN algorithm and outlier detection method.

II. LITERATURE REVIEW

Abhinav Srivastava et al [1] the author uses the ranges of transaction amount as an attribute in the HMM. The author has suggested method for finding the spending profile of cardholders. It is also discussed how the HMM can identify the fraudulent transactions. The simulation results show the advantages of using HMM and learning the profile of the cardholder plays an important role in analyzing fraudulent cases. The result also shows that 80% of the results are accurate and the system is scalable for large data set as well.

Divya.Iyer et al [2] the author uses Hidden Markov Model (HMM) to detect credit card transaction frauds. The training set is tuned with the normal behavior of the card holder. So if credit card transaction is rejected by the trained HMM then that transaction is said to be fraudulent. Care is to be taken that valid and genuine transactions are not considered as fraud. The author also compares various methods with the proposed methods to prove that HMM is much preferred than the other methods.

K.RamaKalyani et al [3] creates a test data and through which the fraudulent activities are detected. This algorithm is also called as an optimization technique based on genetic and natural selection in high computational problems. The author proposes a method to detect credit card fraud and the results are validated using principles of this algorithm. The purpose of detecting fraud cases is to declare it to the client and the service provider.

Renu et al [4] proposed a fraud detection method which involves monitoring the activities of populations to observe and predict undesirable behavior. Undesirable behavior is a set of several habits like intrusion, fraud, delinquency and defaulting.

This research speaks on several credit card fraud detection and telecommunication fraud and different techniques which help in resolving the discussed problems.

Venkata Ratnam Ganji et al [5] the author uses concept of data stream outlier detection algorithm which is based on anti k-nearest neighbors for credit card fraud identification. Whereas traditional methods need to scan the database many times to find the fraudulent transaction, which is not suitable for data stream surroundings. This method makes easier to stop fraudulent transaction happens by Lost and stolen card and Credit card validation checks and detects errors in a sequence of numbers which also helps to detect valid and invalid numbers easily.

III. PROBLEM STATEMENT

Classification consists of assigning a class label into a set of unclassified cases [11].Classification schemes are generally performed on the two approaches namely: statistical and structural [11]. Statistical classification is performed on statistical characterizations of patterns which are generated using probabilistic system; whereas structural classification

depends on the structural interrelationships of features [5]. Supervised classifications are a process in which we gather the information of interest from the dataset and consider it as training data. The statistical classification of each training set is obtained. After the statistical classification is performed the best match from the training data is considered and the class it resembles is chosen. In unsupervised classification, large dataset is considered and divided into several parts or groups based on the natural grouping present in the dataset. Unsupervised classification does not require any labeling of data, all it needs is just that the data within a class should be as close as possible in the measurement space and its distance with other classes should be as high as possible. The purpose of this paper is to study the implementation of K-Nearest Neighbor (KNN) and outlier detection performance results of the same when it is applied on credit card approval system.

IV. CREDIT CARD FRAUD DETECTION

Credit card fraud detection is considered highly confidential and not disclosed to public. Hence, we have discussed a few available methods below.

Table I. Comparison of Various Techniques

Fraud Detection Techniques	Advantage	Disadvantage
Logistic Regression	It produces a simple probability formula for classification. It works well with linear data for credit card fraud detection.	1. It cannot be applied on non-linear data. 2. It is not capable of handling fraud detection at the time of transaction.
Decision Tree	This method can handle non-linear data as well.	1. It involves complex algorithm and even a small change in the data can distract the structure of the tree. Choosing splitting criteria is also complex. 2. It cannot detect fraud at the time of transaction.
Artificial Neural Network	This method is capable of detecting the fraudulent activity at the time of transaction.	1. The number of parameters is to be set before the training is initiated. No rules as such are in place to set the parameters. 2. The network depends on the interconnection between the neurons. Till date there are no methods to determine the optimal topology for a given problem.
Hidden Markov Model.	This method is capable of detecting the fraudulent activity at the time of transaction. The HMM based models reduce the False Positive (FP) transactions predict as fraud though they are really genuine customer.	1. It cannot detect the fraud in initial few transactions.
Support vector machine	This method is capable of detecting the fraudulent activity at the time of transaction.	1. Sometimes it fails to detect fraud cases.
K-Nearest Neighbor Algorithm	There is no requirement of predictive model before classification.	1. The accuracy of the method depends on the measure of distance. 2. It cannot detect the fraud at the time of transaction.

V. FRAUD DETECTION METHODS

Fraud detection is a highly complex function to be performed where there is no system which can guarantee a 100% satisfaction result rate [5]. All the existing methods can likely predict fraud transactions and not assure you about the results. Let’s consider the properties of good fraud detection method:

- a) It must be able to identify the frauds accurately.
- b) It must quickly detect fraud cases.
- c) At any case a genuine transaction should not be considered as fraud.

VI. K-NEAREST NEIGHBOR ALGORITHM

K-nearest neighbor algorithm is used largely in detection systems. It is also proved that KNN works extremely well in credit card fraud detection systems using supervised learning techniques. In this method the new instance query will be classified depending on the KNN category. This method was first utilized by Aha, Albert and Kibler in the year 1991. The results of KNN depend on the below three factors:

- a) The distance metric used to decide the nearest neighbors.
- b) The distance rule that is used for the classification from K-nearest neighbor.
- c) The number of neighbors considered to classify the new sample.

When we study on various credit card fraud detection methods based on supervised statistical pattern recognition, KNN achieves high performance rate without using the priori assumptions about the distributions. The KNN based credit card fraud detection techniques need two major things to be estimated namely the distance or similarity measure between two data instances. In KNN any incoming transaction will be calculated for its nearest point to new incoming transaction. So if the incoming transaction is fraudulent then the algorithm indicates it to be fraud. Larger K values help in reducing the noise in the data set. The distance between two data instances can be computed using different methods. In place of continuous attributes, Euclidean distance is used. In case of categorical attributes, an easy matching coefficient is used. In place of multivariate data, the distance is calculated for every attribute and then combined. The KNN algorithm can be optimized using better distance metrics. In this method both legitimate and fraudulent examples are to be fed in order to train the data sets. This method is fast with minimum false alerts.

VII. K-NEAREST NEIGHBOR CLASSIFICATION

In order to perform the classification using KNN let's consider a new object along with several known examples.

Refer the below image for a better understanding on the concept. The query point is in red circle along with plus and minus nearest points. Now our job is to get the outcome of the query point depending on the selected number of nearest neighbors. We are supposed to find if the query will hold plus or minus sign.

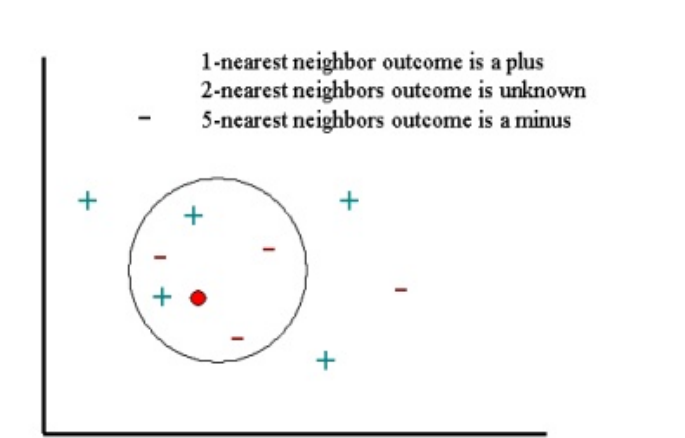


Fig. 1.1 Implementation of KNN

Consider the outcome of KNN for 1 nearest neighbor. So in this case the KNN will provide the query with plus sign. Now increase the number of nearest neighbor to 2. Now there is a plus and minus points close to the query point. Hence both will have equal number of votes. The next step will be considered with 5 KNN points. Now we can define the nearest neighbor point. In this case there are 2 plus and 3 minus signs; hence the outcome will have minus sign.

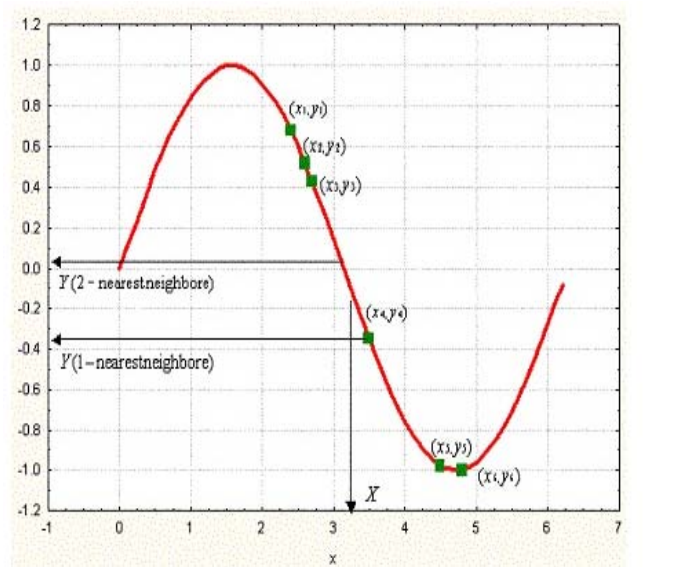


Fig. 1.2 KNN Implementation Graph

VIII. OUTLIER DETECTION

An outlier approach is very different from the traditional observation method. Outlier method is used to detect unusual behaviour of a system using a different mechanism. Outlier detection is classified into two main types based on which labels can be assigned to each group^[6].

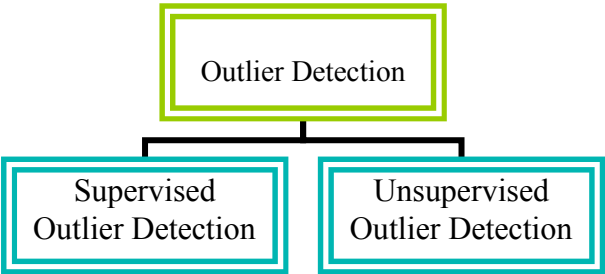


Fig. 2.1 Types of Outlier Detection

A. In *supervised outlier detection* method domain experts model the system to learn and classifier the outlier using training set of data.

B. In *unsupervised outlier detection* objects are clustered into multiple groups based on features. Those objects are far from any group is labeled as outliers.

The below Fig 2.2 shows the difference between unsupervised and supervised outlier detection mechanism.

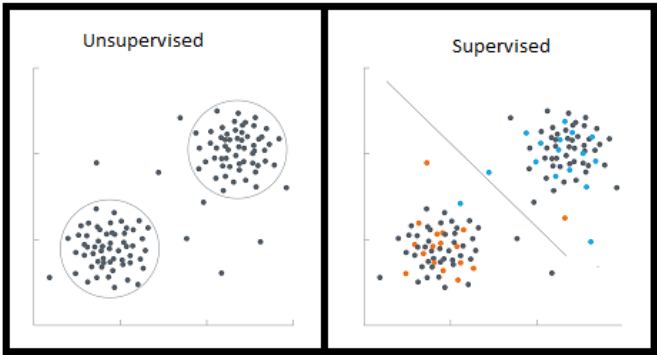


Fig.2.2 Outlier Detection Using Unsupervised and supervised methods

In outlier detection method unsupervised learning is preferred to detect the fraud because it can lead to new explanations and representation of the observation data. Another advantage of using unsupervised data, it does not require prior labeling of data or knowledge about fraudulent methods or transactions. So it need not be trained to discriminate between a legal and illegal transaction. It simply follows the normal behaviour pattern as

an unusual activity or fraudulent. The problem with supervised methods is that the model has to be trained with both fraudulent and non-fraudulent behavior prior to implementing the method in live scenario. Only after the training is performed the system is ready to detect unusual behaviors. The major advantage of using unsupervised method over supervised data is that it need not be trained to discriminate between a legal and illegal transaction.

IX. CONCLUSION

Credit card scam has become much more extensive. To progress safety measures of the monetary transaction systems in a habitual and effectual way, structure a precise and well-organized credit card scam detection system is one of the essential functions for money transactions. By performing over sampling and extracting the principal direction of the data we can use our KNN method to determine the anomaly of the target instance. Hence the KNN method can suit for detecting fraud with the limitation of memory. By the mean time outlier detection mechanism helps to detect the credit card fraud using less memory and computation requirements. Especially outlier detection works fast and well on online large datasets. But compared with power methods and other known anomaly detection methods, experimental results prove that the KNN method is accurate and efficient.

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